

CE902 (2023) — Worked Solutions for Uploaded Section B

Professional Practice and Research Methodology

The uploaded paper explicitly states that **Section A is unavailable**. Therefore this document solves all questions that are actually visible in the file: **Question 18** and **Question 19** in Section B.

Method used throughout: write the governing formula first, substitute the given values carefully, carry the arithmetic through step by step, then interpret the result in plain language.

Question 18

Present value and investment decision

For a future repayment X received after t years, discounted at annual rate r , the present value is:

$$PV = X / (1 + r)^t$$

Because £500 is paid out immediately at time 0, the net present value is:

$$NPV = PV \text{ of repayment} - \text{initial outflow}$$

(a) Compare the offer with an alternative return of 10% per year

Substitute $X = 570$, $r = 0.10$ and $t = 1$:

$$PV = 570 / (1.10)^1$$

$$PV = 518.18$$

$$NPV = 518.18 - 500 = 18.18$$

The NPV is positive. That means the promised £570 in one year is worth more, in present-value terms, than the £500 given up now when the opportunity cost is 10%. So this is a financially acceptable investment at a 10% benchmark.

(b) Recalculate when the best alternative rate is 15% per year

Now use $r = 0.15$ with the same repayment and time period:

$$PV = 570 / (1.15)^1$$

$$PV = 495.65$$

$$NPV = 495.65 - 500 = -4.35$$

Now the NPV is negative. So if 15% can be earned elsewhere, the proposed repayment is not sufficient to compensate for giving up £500 today. The investment decision therefore changes: under a 15% alternative, the offer should be rejected.

Question 19

(a) Cash flow and the role of the cash flow statement

Cash flow is the movement of cash into and out of a business over a period of time. Cash inflows can come from sales, borrowing, or asset disposals, while cash outflows can arise from wages, materials, rent, loan repayments, tax, and capital expenditure.

Cash flow statements are important because they show whether the business can actually meet its day-to-day obligations in cash. A firm may report accounting profit and still face liquidity problems if cash arrives too late or payments are badly timed. The statement therefore helps managers, lenders, and

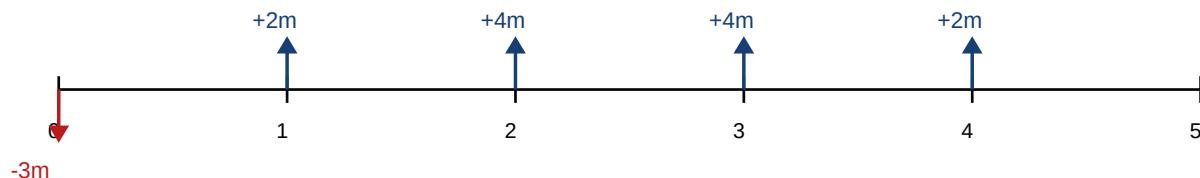
investors assess liquidity, solvency, timing of receipts and payments, and the sustainability of operations.

(b) Discounted cash flow appraisal of Projects A and B

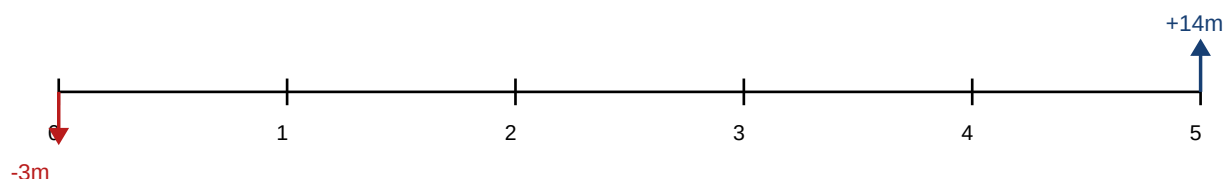
The company discounts all future cash flows at 9%, because the initial funds are raised at that rate. For each year t , the discounted cash flow is:

$$\text{Discounted cash flow at year } t = \text{Cash flow}_t / (1.09)^t$$

Project A cash-flow timeline



Project B cash-flow timeline



Year	Project A cash flow	Project A discounted value	Project B cash flow	Project B discounted value
0	-\$3.00m	\$-3.000m	-\$3.00m	\$-3.000m
1	\$2.00m	\$1.835m	\$0.00m	\$0.000m
2	\$4.00m	\$3.367m	\$0.00m	\$0.000m
3	\$4.00m	\$3.089m	\$0.00m	\$0.000m
4	\$2.00m	\$1.417m	\$0.00m	\$0.000m
5	\$0.00m	\$0.000m	\$14.00m	\$9.099m

Step 1: discounted cash flows year by year

Project A, year 1: $PV = 2 / (1.09)^1 = 1.835$ million dollars.

Project A, year 2: $PV = 4 / (1.09)^2 = 3.367$ million dollars.

Project A, year 3: $PV = 4 / (1.09)^3 = 3.089$ million dollars.

Project A, year 4: $PV = 2 / (1.09)^4 = 1.417$ million dollars.

Project B, year 5: $PV = 14 / (1.09)^5 = 9.099$ million dollars.

Step 2: net present value of each project

$$NPV(\text{Project A}) = -3.000 + 1.835 + 3.367 + 3.089 + 1.417 + 0.000$$

$$NPV(\text{Project A}) = 6.707 \text{ million dollars}$$

$$NPV(\text{Project B}) = -3.000 + 0 + 0 + 0 + 0 + 9.099$$

$$NPV(\text{Project B}) = 6.099 \text{ million dollars}$$

Both projects have positive NPVs, so both are financially worthwhile relative to the 9% discount rate. However, Project A has the larger NPV.

Step 3: recommendation

Because Project A has an NPV of about \$6.707m, while Project B has an NPV of about \$6.099m, the analyst should recommend **Project A** if the two projects truly carry the same risk. It creates the larger increase in shareholder value after allowing for the time value of money.

(iv) Purpose of the three main annual financial statements

Statement	Purpose
Balance Sheet	Shows the company's financial position at a point in time: what it owns (assets), what it owes (liabilities), and the residual equity of the owners.
Profit & Loss Account	Shows financial performance over a period: revenue earned, expenses incurred, and the resulting profit or loss.
Cash Flow Statement	Shows the actual movement of cash through operating, investing, and financing activities, revealing liquidity and the ability to meet cash obligations.

End of worked solutions for all questions visible in the uploaded CE902 (2023) file.